

Stat230-F26 | Quiz 1

Name:

For open-ended questions, you must show all work and/or explain your steps in order to get full credit or be considered for partial credit. You do not need to simplify any quantities (eg leave sums as sums and fractions as fractions) but you can use a calculator to do so if you find it helpful.

For any True/False, multiple choice, or fill-in-the-blank questions, you do not need to justify your answer unless it's explicitly asked.

You are not allowed to communicate or share anything about this quiz with anyone besides the professor until graded quizzes are returned to you.

I certify that all work is my own and that I will not discuss the content of this quiz with anyone in or outside of class, and that I will not share it, upload it, or input its questions into an AI prompt.

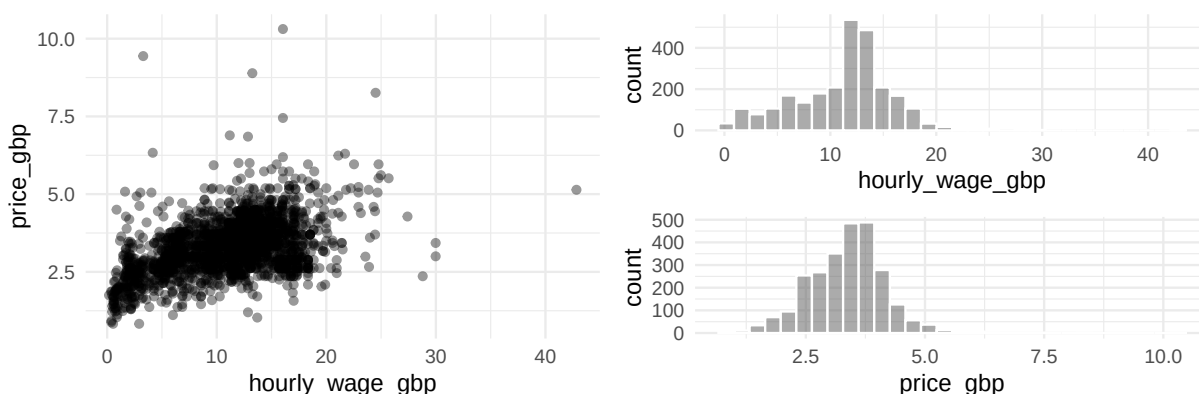
Signature: _____

We will investigate the “Cappuccino Index” – how a cappuccino’s price compares to the local hourly wage, across cafés around the world. Each observation is an individual café. The variables include:

- **country:** the country the café is located in
- **city:** the city the café is located in
- **hourly_wage_gbp:** the local hourly wage, converted to British pounds (GBP)
- **price_gbp:** the price of a cappuccino at that café, converted to British pounds (GBP)

The first few rows of the dataset are shown below. You also have some EDA and output from a linear regression model on this page.

country	city	hourly_wage_gbp	price_gbp
Denmark	Copenhagen	16.04	10.31
China	Shenzhen	3.29	9.44
USA	West Richland	13.25	8.89



Call:

```
lm(formula = price_gbp ~ hourly_wage_gbp, data = cafe)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.5587	-0.4064	0.0200	0.3962	6.7446

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.413137	0.037788	63.86	<2e-16 ***
hourly_wage_gbp	0.085805	0.003109	27.60	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7248 on 2592 degrees of freedom

Multiple R-squared: 0.2272, Adjusted R-squared: 0.2269

F-statistic: 761.8 on 1 and 2592 DF, p-value: $< 2.2\text{e-}16$

1. Write out the *theoretical* SLR model. Make sure to define Y and X in context, include all parameters, and state the assumption placed on the error term.

2. Write out the *fitted* SLR model.

3. Interpret the *intercept* estimate from the summary in context.

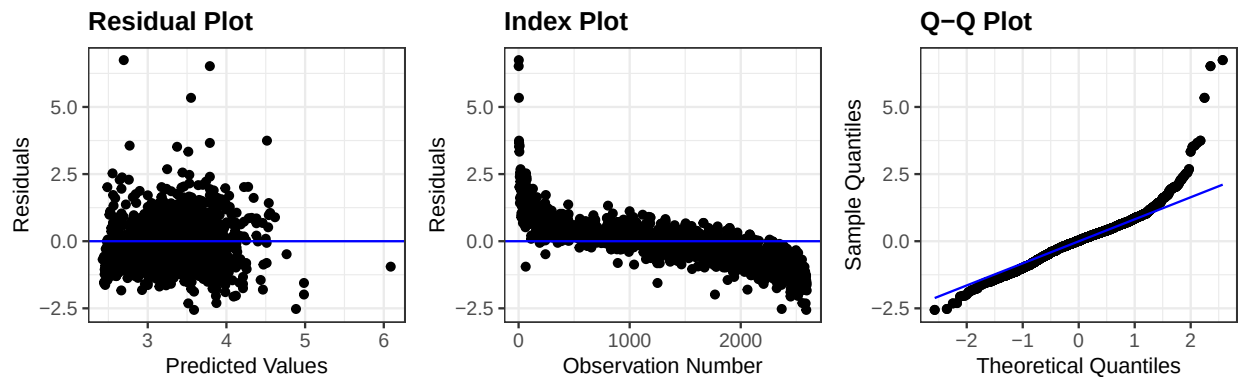
4. (a) Find a 90% confidence interval for the slope. For full credit, include the mathematical formula, the formula with appropriate numbers plugged in, and your final interval. If you do not know a quantity that you need, explain how you would find it. (In addition to the `summary()` output, there are some critical values on the last page of the quiz.)

- (b) Provide an interpretation of your interval in (a) in context.

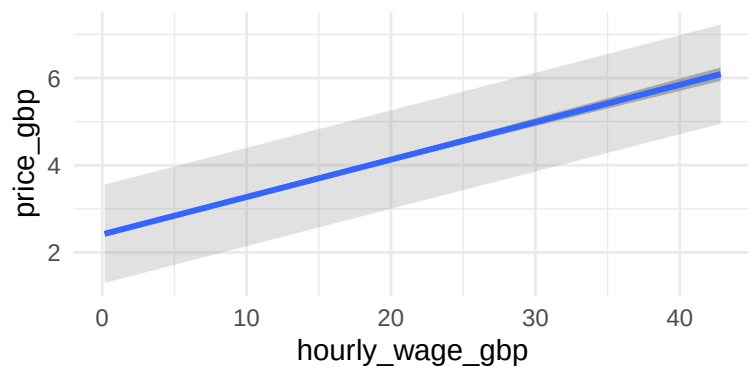
5. Based on the `summary()` output for the `cafe_lm` model, which of the following provides the best interpretation of the reported residual standard error?
- ☐ (a) For every £1 increase in a café's local hourly wage, the predicted cappuccino price increases by that amount
 - ☐ (b) When using this model to predict a café's cappuccino price based on the local hourly wage, actual prices typically deviate from the predicted prices by about that amount
 - ☐ (c) The linear model using hourly wage explains that percentage of the overall variation in cappuccino prices
 - ☐ (d) The sample slope estimate is expected to vary by that amount from sample to sample due to natural sampling variability
 - ☐ (e) The average cappuccino price across all cafés in this dataset is that amount
 - ☐ (f) The test statistic (t) for the slope is that amount
6. (a) Write out the null and alternative hypotheses that the p-value reported for `hourly_wage_gbp` in the summary output corresponds to.

(b) Based on this p-value, what is your conclusion in the context of the problem?

7. For this question, use the regression output and residual plots and circle TRUE or FALSE for each part.
- (a) TRUE or FALSE: The linearity condition is **not met** because the R^2 value is small and there is no obvious linear trend in the original scatterplot
 - (b) TRUE or FALSE: The normality condition is **not met** because the Q-Q plot departs from the diagonal line in the upper tail
 - (c) TRUE or FALSE: The independence condition is **met** because the Index plot looks completely random
 - (d) TRUE or FALSE: The normality condition is **not met** because the histogram of the predictor variable (X) is right-skewed
 - (e) TRUE or FALSE: In the ideal SLR scenario, all of the points in the Residual plot will fall exactly on the horizontal line shown
 - (f) TRUE or FALSE: The independence condition is often the hardest to check using only residual plots because it usually requires knowing how the data was collected
 - (g) TRUE or FALSE: The equal variance condition is the most important for SLR
 - (h) TRUE or FALSE: If any of the conditions are violated, we cannot use the SLR model to say anything about the association between X and Y



8. Using the graph below, report the appropriate interval (approximations are fine!) for each quantity.



- (a) The projected cappuccino price at one specific café in a city with an hourly wage of £15.
- (b) The average cappuccino price across all cafés in cities with an hourly wage of £10.
- (c) A café in a city where the hourly wage is only £1.

9. Show how the following interval was computed using the appropriate mathematical formula. You should report the formula and plug in all numbers, but do not need to simplify. If you do not know a quantity, explain how you would find it.

```
predict(cafe_lm, newdata = data.frame(hourly_wage_gbp = 15),
        interval = "prediction", level = .88, se.fit = TRUE)
```

```
$fit
      fit      lwr      upr
1 3.700218 2.572547 4.827889
```

```
$se.fit
[1] 0.01837647
```

```
$df
[1] 2592
```

```
$residual.scale
[1] 0.724824
```

10. Answer the following parts using the interval from Q9 as a baseline. For each part, assume nothing else changes but what is specified.

(a) If we instead built the interval for a café with an hourly wage of £30, would the interval be:

- ☐ wider
- ☐ narrower
- ☐ the same
- ☐ not enough information to tell

(b) If there were an additional 500 cafés in the dataset, would the interval be:

- ☐ wider
- ☐ narrower
- ☐ the same
- ☐ not enough information to tell

(c) If the standard deviation of `price_gbp` *increased*, would the interval be:

- ☐ wider
- ☐ narrower
- ☐ the same
- ☐ not enough information to tell

(d) If the standard deviation of `hourly_wage_gbp` *decreased*, would the interval be:

- ☐ wider
- ☐ narrower
- ☐ the same
- ☐ not enough information to tell

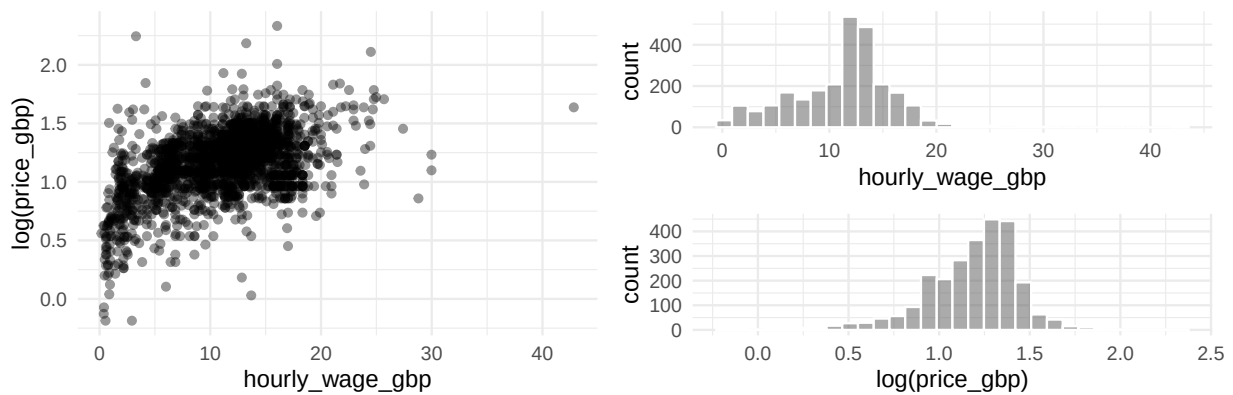
11. I remember from class that a log transformation can help with skew, so I decide to try taking the log of `price_gbp`. The graphs below show some EDA, linear regression output, and residual plots.

(a) Write out the fitted SLR equation.

(b) I look back at my class notes and conclude “*For every additional £1 in hourly wage, the average cappuccino price is expected to increase by [the reported slope] pounds.*” This interpretation is incorrect. Explain why in 1 sentence or less, then provide a correction.

(c) I also conclude “*The model explains [the reported R^2 as a percent] of the overall variability in cappuccino prices.*” This interpretation is incorrect. Explain why in 1 sentence or less, then provide a correction.

(d) Which model do you prefer in this case? Please answer in 2 sentences or less and use one graphical or numeric output to justify your answer.



Call:

```
lm(formula = log(price_gbp) ~ hourly_wage_gbp, data = cafe)
```

Residuals:

Min	1Q	Median	3Q	Max
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-1.22670 -0.10686 0.03171 0.13554 1.28853

Coefficients:

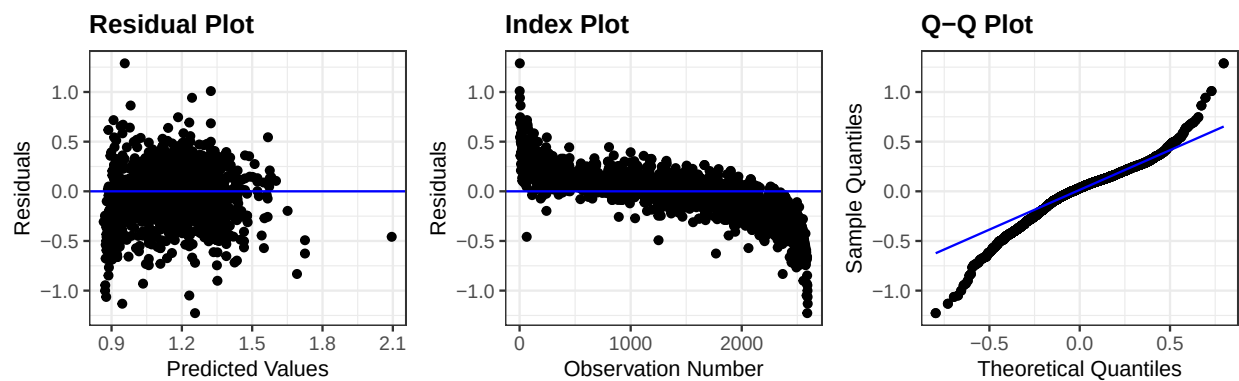
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.8616706	0.0117138	73.56	<2e-16 ***
hourly_wage_gbp	0.0288021	0.0009637	29.89	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2247 on 2592 degrees of freedom

Multiple R-squared: 0.2563, Adjusted R-squared: 0.256

F-statistic: 893.3 on 1 and 2592 DF, p-value: < 2.2e-16



```
n <- nrow(cafe)
qt(.01, df = n - 2)
```

```
[1] -2.327787
```

```
qt(.025, df = n - 2)
```

```
[1] -1.96088
```

```
qt(.05, df = n - 2)
```

```
[1] -1.645442
```

```
qt(.06, df = n - 2)
```

```
[1] -1.555286
```

```
qt(.1, df = n - 2)
```

```
[1] -1.281878
```

```
qt(.12, df = n - 2)
```

```
[1] -1.175257
```